# Hao Xu

haxu0087@uni.sydney.edu.au • Google Scholar • LinkedIn • GitHub

## Research Interests

Foundation models for decision-making • Partially Observable Markov Decision Processes (POMDPs) • LLM applications in manufacturing and materials science

## Education

**Ph.D. in Mathematics and Statistics** *2025–Present*
*The University of Sydney, Sydney, Australia*
*Research:* Foundation models for decision-making, POMDPs, and AI/LLM applications in manufacturing and materials science.

**MPhil in Smart Manufacturing** *2022–2024*
*HKUST (Guangzhou), Guangzhou, China*
*Research:* Anomaly detection in industrial manufacturing; high-dimensional data processing for point clouds and images.

**M.S. in Data Science** *2021–2022*
*City University of Hong Kong, Hong Kong, China*
*Research:* Deep learning and predictive modeling; attention-based financial sentiment analysis and epidemiological mortality monitoring.

**B.Eng. in Materials Science and Engineering** *2016–2020*
*Chongqing University, Chongqing, China*
*Research:* Materials synthesis, characterization, and failure analysis; ODS steels, Mg-based hydrogen storage alloys, and high-voltage testing. **Grade Rank: Top 5%.**

## Publications

** indicates co-first authors; † indicates corresponding authors.*

[1] Xiangru Jian*, **Hao Xu*†**, Wei Pang*, Xinjian Zhao, Chengyu Tao, Qixin Zhang, Xikun Zhang†, Chao Zhang, Guanzhi Deng, Jingxi Xue, Juan Du, Tianshu Yu, Garth Tarr, Linqi Song, Qiuzhuang Sun, Dacheng Tao.
**FORGE: Fine-grained Multimodal Evaluation for Manufacturing Quality Assurance.**
*Findings of the Association for Computational Linguistics: EMNLP 2026.* [Paper]

[2] **Hao Xu***, Xiangru Jian*†, Xinjian Zhao*, Wei Pang*, Chao Zhang, Suyuchen Wang, Qixin Zhang, Zhengyuan Dong, Joao Monteiro, Bang Liu, Qiuzhuang Sun, Tianshu Yu†.
**GraphOmni: A Comprehensive and Extendable Benchmark Framework for Large Language Models on Graph-theoretic Tasks.**
*International Conference on Learning Representations (ICLR)*, 2026. [arXiv]

[3] Chengyu Tao, **Hao Xu**, Juan Du†.
**F2PAD: A General Optimization Framework for Feature-Level to Pixel-Level Anomaly Detection.**
*arXiv preprint arXiv:2407.06519*, 2024. [arXiv]

[4] **Hao Xu**, Juan Du†, A. Wang, Y.C. Chen.
**Ano-SuPs: Multisize Anomaly Detection for Manufactured Products by Identifying Suspected Patches via Vision Transformer.**
*INFORMS Journal on Data Science*, 2026. [Paper]

## Invited Talks

- **From AI for Manufacturing to LLMs for Manufacturing: Fine-grained Multimodal Evaluation for Manufacturing Scenarios.**
  Invited lecture to the SJTU MEM Summer Study Tour delegation, The University of Sydney, Sydney, Australia, Aug 2026.

## Selected Awards

Finalist, Best Paper Competition — INFORMS Conference on Quality, Statistics, and Reliability *2025*
Outstanding Graduate of Chongqing City *2020*
National College Students Summer Social Practice Special Action Bronze Award *(Deputy Leader)* *2018*

## Academic Service

- **Conference Reviewer:** International Conference on Learning Representations (ICLR); Conference on Neural Information Processing Systems (NeurIPS); Conference on Language Modeling (COLM)